

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

*Assessment Tool Weightage: 30% Software: Cisco
Packet Tracer, Wireshark*

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. IPv4 Network Configuration Debugging

Given Configuration

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

The workstation cannot communicate with systems in **192.168.10.0/26**.

Step 1: Identify the Subnet

Subnet Mask = /26

A /26 subnet has a block size of **64**.

Subnets are:

Subnet	Address Range
192.168.10.0/26	192.168.10.0 – 192.168.10.63
192.168.10.64/26	192.168.10.64 – 192.168.10.127
192.168.10.128/26	192.168.10.128 – 192.168.10.191
192.168.10.192/26	192.168.10.192 – 192.168.10.255

The configured IP **192.168.10.65** belongs to the **192.168.10.64/26** subnet.

Step 2: Network Details

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host Range:** 192.168.10.65 – 192.168.10.126

Step 3: Verify IP Address and Gateway

- IP Address **192.168.10.65** → Valid ✓
- Gateway **192.168.10.1** → Invalid ✕

Reason:

The gateway belongs to subnet **192.168.10.0/26**, while the workstation belongs to **192.168.10.64/26**.

Devices and their gateway must be in the **same subnet**.

Step 4: Configuration Errors

1. Workstation is configured in the wrong subnet.
2. Default gateway belongs to a different subnet.
3. Communication fails because packets cannot reach the gateway.

Step 5: Optimized Configuration

If the workstation should belong to **192.168.10.0/26**, configure:

- **IP Address:** 192.168.10.2
- **Subnet Mask:** 255.255.255.192
- **Default Gateway:** 192.168.10.1

OR

If it should remain in **192.168.10.64/26**, configure:

- **IP Address:** 192.168.10.65
- **Default Gateway:** 192.168.10.65 or another router interface such as **192.168.10.66** configured for that subnet.

Step 6: Usable Hosts

For /26

Total addresses = 64

Usable hosts = **64 – 2 = 62**

Final Answer

Item	Value
Configured Subnet	192.168.10.64/26
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Host Range	192.168.10.65 – 192.168.10.126
Gateway Valid?	No
Optimized IP	192.168.10.2
Optimized Gateway	192.168.10.1
Usable Hosts	62

2. IPv6 Debugging

Given

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Step 1: Expand Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

Using /64

Device A

Network Prefix

2001:db8:0:0::/64

Device B

Network Prefix

2001:db8:0:0::/64

Step 3: Same Subnet?

Yes.

The first 64 bits are identical.

Both belong to

2001:db8:0:0::/64

Step 4: Why Communication Fails?

Possible reasons:

- Incorrect default gateway
- Firewall blocking traffic
- Neighbor Discovery (NDP) issue
- Duplicate IPv6 address
- Interface disabled

There is **no prefix mismatch**.

Step 5: Optimized Configuration

Device A
2001:db8::10/64
Device B
2001:db8::20/64
Gateway
2001:db8::1

Step 6: Host Addresses

Host bits = 64
Available hosts
 $2^{64} = 18,446,744,073,709,551,61$

Final Answer

Item	Device A	Device B
Expanded Address	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000:0001:0000:0000:0001
Network Prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Same Subnet?	Yes	Yes

Available Hosts
 2^{62}

3. IPv4 Subnet Optimization Using VLSM

Given
Network
192.168.1.0/24
Subnets
• /26
• /27
• /28

Existing Allocation

Subnet 1 (/26)
Network
192.168.1.0
Host Range
192.168.1.1 – 192.168.1.62
Broadcast
192.168.1.63
Usable Hosts
62

Subnet 2 (/27)
Network
192.168.1.64
Host Range

192.168.1.65 – 192.168.1.94

Broadcast

192.168.1.95

Usable Hosts

30

==

Subnet 3 (/28)

Network

192.168.1.96

Host Range

192.168.1.97 – 192.168.1.110

Broadcast

192.168.1.111

Usable Hosts

14

Inefficient Allocation

- Small departments receiving /26 waste many IPs.
- Large departments using /28 run out of addresses.

Addressing Mistakes

- Incorrect subnet size assigned.
- Overlapping subnets.
- Wrong host capacity planning.

Optimized VLSM Scheme

Assign the largest department first.

Example:

Department	Required Hosts	Recommended Subnet
Dept A	60	/26
Dept B	25	/27
Dept C	12	/28

This minimizes unused addresses.

4. IPv6 Debugging

Answer: Same as **Question 2**.

- Expanded Address:
 - Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329
 - Device B: 2001:0db8:0000:0000:0001:0000:0000:0001
- Network Prefix: **2001:db8:0:0::/64**
- Same Subnet: **Yes**
- Issue: Not an addressing mismatch; likely gateway, NDP, firewall, or interface problem.
- Optimized Addresses:
 - Device A: 2001:db8::10/64
 - Device B: 2001:db8::20/64
 - Gateway: 2001:db8::1
- Available Hosts: **2⁶⁴ – 2 = 18,446,744,073,709,551,616**

5. Routing Table Debugging
Given Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP

192.168.2.100

Step 1: Longest Prefix Match

Destination

192.168.2.100

Matches

192.168.2.0/24

Step 2: Next-Hop Router

Packets are forwarded to

10.0.0.3

Step 3: Verify Destination

192.168.2.100 belongs to

- Network = **192.168.2.0**
- Broadcast = **192.168.2.255**

Therefore,

Yes, it belongs to the intended subnet.

Step 4: Routing Errors

Possible causes:

- Incorrect next-hop IP.
- Down interface.
- Missing route on next router.
- Destination host offline.
- Incorrect subnet mask.

Step 5: Optimized Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Ensure that:

- Interfaces are active.
- Next-hop routers are reachable.
- Routing entries are verified.

Step 6: If No Route Matches

The router uses the

Default Route (0.0.0.0/0)

and forwards packets to

10.0.0.1